A MAGIS-100 Experimental Forecast for Momentum-Quadratic Dephasing

A Short Technical Note on Experimental Discriminants for the Fixed-Cell Transport Signature

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Abstract

This note translates the momentum-quadratic dephasing prediction of the fixed-cell transport model into an experimentally oriented forecast for MAGIS-100. The goal is not to rederive the underlying theory, but to express the predicted signature in the variables most relevant to a long-baseline clock atom interferometer and to identify concrete control tests that distinguish it from standard technical loss channels. The central prediction is a residual dephasing component whose leading dependence is quadratic in large-momentum-transfer order, linear in interrogation time, and stronger for smaller pre-split coherence length or, under the thermal identification, for higher source temperature. The sharpest content of the forecast is therefore the joint scaling law across multiple independent experimental control variables, rather than the absolute value of its overall coefficient, which still depends on parameters not yet derived microscopically. The note proposes a compact discrimination program built from LMT sweeps, source-preparation sweeps, interrogation-time sweeps, and ordinary pulse recalibration. Its main experimental value is not merely a positive forecast, but a falsifiable strategy for separating a substrate-driven coherence burden from ordinary technical contrast loss.

1 Purpose and scope

MAGIS-100 is designed to operate as a long-baseline clock atom interferometer in a regime where large momentum transfer (LMT), long free-fall times, and large arm separations are all central design features [1]. That makes it a natural target for testing whether contrast degradation contains a contribution beyond ordinary technical pulse inefficiency or classical noise.

The present note is therefore intentionally narrow. It does not attempt to restate the full NDK framework or the derivation of emergent Schrödinger dynamics. Those tasks are handled in the upstream papers [2, 3]. Instead, the goal here is to provide an experimentally oriented forecast based on the LMT dephasing Letter [4]:

(i) express the predicted signature in observable interferometer variables;

(ii) state how it differs from standard LMT loss channels;

(iii) identify simple sweeps that could support or disfavor the effect.

The sharpest content of the present forecast is the functional form of the residual scaling law rather than the absolute value of its overall coefficient. The note predicts a joint dependence on

LMT order, interrogation time, and source preparation that is not shared by standard technical loss channels. By contrast, the absolute normalization still depends on parameters that have not yet been derived microscopically, so the primary experimental discriminator is the shape of the residual law across controlled sweeps rather than a single headline magnitude prediction.

MAGIS-100 remains under active construction and development as a long-baseline clock atom interferometer and pathfinder instrument [1, 5].

2 Prediction in experimental variables

2.1 Residual dephasing law in LMT-letter notation

After standard technical-noise and environmental modeling, define the residual visibility by

$$V_{\mathrm{res}} \equiv \frac{\mathcal{C}_{\mathrm{obs}}}{\mathcal{C}_{\mathrm{tech}}}, \tag{1}$$

and the associated residual dephasing exponent by

$$D_{\mathrm{pred}} \equiv -\ln V_{\mathrm{res}}. \tag{2}$$

In the fixed-cell transport bridge of [4], the cell-level transport burden is

$$\mathcal{D}_{\mathrm{tr}} \equiv \pi\,\chi\,\mathcal{I}_*/A\,c\,\ell_{\mathrm{P}}\,T\,n^2 k_{\mathrm{eff}}^2, \tag{3}$$

where T is the interrogation time, n is the effective LMT order, k_{eff} is the effective wavevector, χ is an order-unity geometric factor, and $\mathcal{I}_*/A$ is the adopted directional-strain benchmark.

The operational coherence-cell area is taken to be

$$A_* = \eta\,\lambda_{\mathrm{coh},0}^2, \tag{4}$$

with η a geometry-convention factor and $\lambda_{\mathrm{coh},0}$ the pre-split coherence length of the atomic source. A recent primitive-cell closure analysis strengthens this ansatz by showing that the underlying NDK support cell is naturally modeled as a finite disk-like lateral footprint rather than as an arbitrary order-unity patch [6]. In that sense, η should not be interpreted as a wholly free nuisance parameter, but as the factor that records how the laboratory coherence-length convention is mapped onto an effective primitive cell area. The present note does not claim that this laboratory mapping has already been derived microscopically. It only uses the closure result to motivate a sharper geometric interpretation of the fixed-cell bridge. The predicted residual dephasing is then

$$D_{\mathrm{pred}} \equiv \mathcal{D}_{\mathrm{tr}}\,\frac{w_{\mathrm{coh}}^2}{A_*} = \frac{\pi\chi}{\eta}\,\mathcal{I}_*/A\,c\,\ell_{\mathrm{P}}\,T\,n^2 k_{\mathrm{eff}}^2\left(\frac{w_{\mathrm{coh}}}{\lambda_{\mathrm{coh},0}}\right)^2, \tag{5}$$

where w_{coh} is the transverse coherence-support width relevant to the evolving cloud.

At this stage of the program, the most robust content of the forecast is the shape of the residual scaling law rather than the absolute value of its full normalization. The latter still depends on several parameters that are not yet derived microscopically in a laboratory reduction, including the geometric transfer factor χ, the thermal bridge factor ξ_{th}, the adopted benchmark $\mathcal{I}_*/A$, and experimental coherence-transfer quantities such as w_{coh}. The present note therefore treats the overall coefficient as a phenomenological transfer quantity, while taking the joint dependence on n^2, T, and source preparation as the sharpest experimental prediction.

2.2 Temperature scaling under a thermal coherence identification

The fixed-cell ansatz assumes that the pre-split coherence length is set by thermodynamic preparation. As a working identification, take

$$\lambda_{\mathrm{coh},0}^2 = \xi_{\mathrm{th}}^2\, \lambda_{\mathrm{th}}^2, \qquad \lambda_{\mathrm{th}}^2 = \frac{2\pi\hbar^2}{m\, k_B \Theta}, \tag{6}$$

where Θ is the source temperature, m is the atomic mass, and ξ_{th} is an order-unity bridge factor.

Substituting equation (6) into equation (5) gives

$$D_{\mathrm{pred}}(\Theta) = \frac{\chi}{2\eta\, \xi_{\mathrm{th}}^2}\, \mathcal{I}_*/A\, c\, \ell_{\mathrm{P}}\, T\, n^2 k_{\mathrm{eff}}^2\, w_{\mathrm{coh}}^2\, \frac{m\, k_B \Theta}{\hbar^2}. \tag{7}$$

Thus, in the fit-law language

$$D_{\mathrm{pred}}(n) = a_0 + a_1 n^2, \tag{8}$$

the quadratic dephasing coefficient satisfies

$$a_1(T, \Theta) \propto T\, \Theta \tag{9}$$

at fixed geometry and fixed transverse coherence-support width.

In the forecast figure below, this manifests as a vertical rescaling of the underlying n-dependence. At fixed geometry and controlled transverse coherence-support width, varying the source temperature Θ changes the overall coefficient of the residual law without altering its log-log shape as a function of n.

2.3 What the prediction means operationally

The forecast is therefore not simply "contrast gets worse at larger LMT." Many standard mechanisms do that. The claim is more specific:

(a) the additional loss enters as a dephasing exponent rather than merely a cumulative pulse-population failure;

(b) it scales quadratically with the LMT enhancement variable;

(c) it strengthens with longer interrogation time;

(d) it strengthens when the pre-split coherence length $\lambda_{\mathrm{coh},0}$ is reduced, equivalently when $(w_{\mathrm{coh}}/\lambda_{\mathrm{coh},0})^2$ increases.

Those four features, taken together, are the real experimental signature.

In the plotted forecast, the remaining shaded spread should be read as a model-side sensitivity band generated by the residual sweep over $(w_{\mathrm{coh}}, \lambda_{\mathrm{coh},0})$, not as a statistical error bar or an experimental uncertainty estimate.

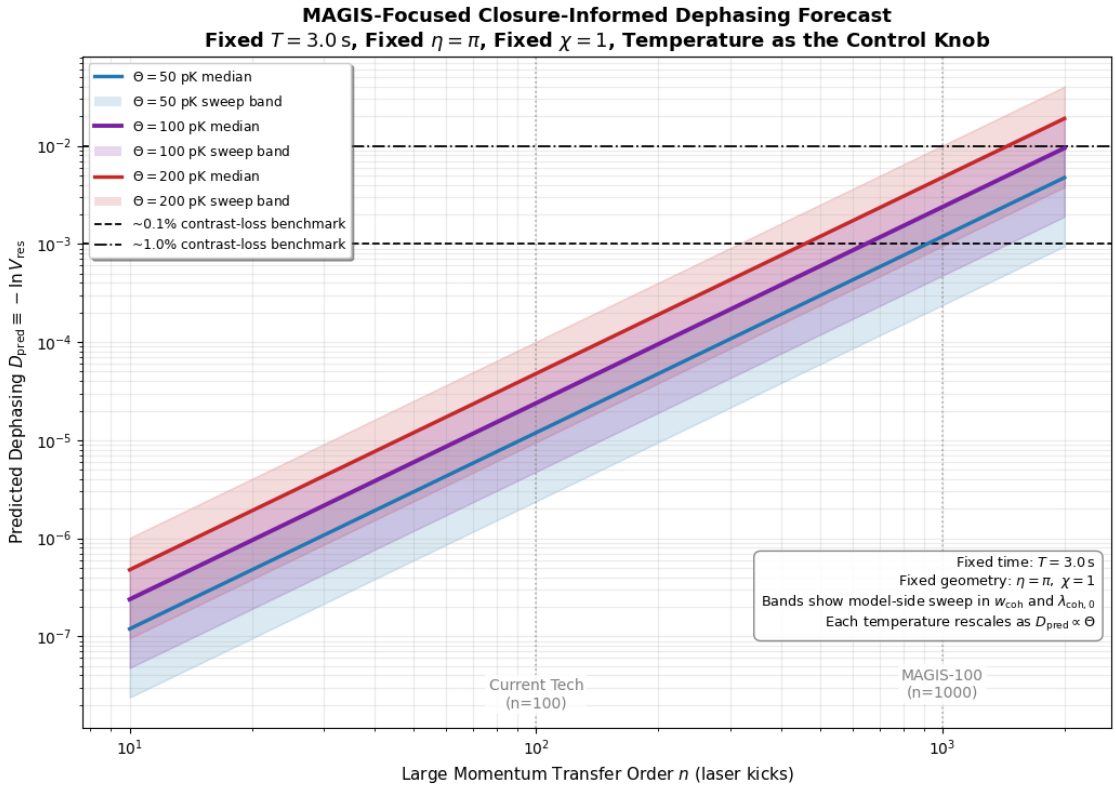

Figure 1: **MAGIS-focused closure-informed dephasing forecast with temperature as the control knob.** The plotted forecast fixes $T = 3.0\,\mathrm{s}$ and uses a closure-informed disk-cell geometry, with $\eta = \pi$ under the convention that the pre-split coherence length $\lambda_{\mathrm{coh},0}$ is mapped to an effective cell radius [6]. The blue, purple, and red curves show the median predicted residual dephasing for representative source temperatures $\Theta = 50\,\mathrm{pK}$, $100\,\mathrm{pK}$, and $200\,\mathrm{pK}$, respectively. The shaded bands show the corresponding model-side coherence-parameter sensitivity range obtained from the remaining sweep in $(w_{\mathrm{coh}}, \lambda_{\mathrm{coh},0})$; they represent uncertainty within the present phenomenological forecast, not an experimental confidence interval. The horizontal black lines indicate approximate 0.1% and 1% contrast-loss benchmarks. Because the temperature dependence enters here as a multiplicative rescaling at fixed geometry and fixed transverse coherence-support width, the three temperature families have the same log-log shape and are separated vertically by the corresponding temperature ratios. Accordingly, the forecast remains small at current $n \sim 10^2$ scales but enters the sub-percent to percent-level window in the $n \sim 10^3$ regime relevant to MAGIS-100-like architectures under the adopted benchmark assumptions.

3 Why this is not just an ordinary LMT technical loss

Recent work on LMT clock interferometry has emphasized that several apparent scaling concerns are milder than once feared. In particular, sequential state inversions from alternating directions lead to cumulative population error that scales linearly with pulse number rather than quadratically, and laser-frequency noise need not be a practical blocker for high-fidelity LMT clock interferometers [7]. Likewise, broader LMT studies continue to frame imperfect pulse efficiencies, parasitic paths, and operating-regime optimization as central technical limits [8].

That is precisely why the fixed-cell prediction is experimentally interesting: it is not a claim about ordinary pulse fidelity accumulation. The proposed dephasing burden arises from the transported phase-gradient load borne by the split matter-wave branches in the reduced sector, together with the fixed-cell closure used to convert that burden into an effective contrast floor [4, 3]. It

should therefore be tested against, not conflated with, standard pulse-imperfection channels.

A compact way to summarize the distinction is:

$$D_{\text{tech}} = D_{\text{pulse}} + D_{\text{wavefront}} + D_{\text{inhom}} + D_{\text{readout}} + \cdots , \tag{10}$$

whereas the fixed-cell contribution predicts

$$D_{\text{pred}} \propto T\, n^2\, k_{\text{eff}}^2 \left(\frac{w_{\text{coh}}}{\lambda_{\text{coh},0}}\right)^2 . \tag{11}$$

The useful experimental question is therefore not whether total contrast loss exists, but whether a residual component with the scaling of equation (11) survives after technical channels are characterized.

4 A minimal discrimination program for MAGIS-100

4.1 LMT sweep

At fixed source preparation and fixed interrogation time, measure the contrast as a function of the LMT control parameter. The key question is whether the residual dephasing exponent extracted from

$$-\ln\left(\frac{\mathcal{C}_{\text{obs}}}{\mathcal{C}_{\text{tech}}}\right)$$

is better fit by a linear law or a quadratic law in n after standard technical models are accounted for.

A clean quadratic residual would be the first nontrivial support for the fixed-cell signature.

4.2 Source-preparation sweep

At fixed n and fixed T, vary the source preparation in a way that changes the pre-split coherence length $\lambda_{\text{coh},0}$, or, under the thermal identification, the source temperature Θ. The model predicts:

$$D_{\text{pred}} \propto \left(\frac{w_{\text{coh}}}{\lambda_{\text{coh},0}}\right)^2 \qquad \text{or, under the thermal identification,} \qquad D_{\text{pred}} \propto \Theta. \tag{12}$$

This is one of the most important discriminants, because many ordinary pulse-calibration channels are not expected to track the pre-split coherence length $\lambda_{\text{coh},0}$ or source temperature Θ in this direct way.

The temperature sweep is especially valuable because it is already part of the present derived phenomenological bridge, not an external speculative discriminator. By contrast, a possible future orientation-dependent refinement of the experimental geometry factor χ may be worth testing as an auxiliary cross-check, but the present note does not require or derive a quantitative angular law for that effect.

4.3 Interrogation-time sweep

At fixed preparation and fixed n, the fixed-cell model predicts an approximately linear growth with interrogation time,

$$D_{\text{pred}} \propto T. \tag{13}$$

A residual contribution that grows with T while keeping its preparation dependence would support the interpretation of the effect as a genuine branch-burden dephasing process rather than a one-shot pulse artifact.

4.4 Recalibration and pulse-optimization sweep

Technical pulse-loss mechanisms should be sensitive to:

- improved pulse shaping,
- reduced laser-frequency noise,
- better control of parasitic paths,
- operating-point optimization.

If a contrast-loss component remains after those improvements and continues to follow the scaling of equation (11), that residual becomes especially interesting.

Conversely, if the apparent n^2 trend disappears under ordinary pulse optimization, the fixed-cell interpretation is disfavored.

5 A practical discrimination summary

To isolate the proposed signature in the laboratory, the defining behaviors of the geometric coherence burden and the primary competing technical systematics are summarized below.

- **NDK fixed-cell residual dephasing.** The predicted residual exponent is D_{pred}, with leading quadratic dependence on n, linear dependence on interrogation time T, and stronger dephasing for smaller pre-split coherence length $\lambda_{\mathrm{coh},0}$. Under the thermal identification, the same prediction becomes a linear dependence of the quadratic coefficient on source temperature Θ.
- **Ordinary pulse-infidelity accumulation.** This is expected to track pulse calibration and sequence fidelity more directly, and is typically linear in pulse number or otherwise implementation-specific rather than a clean residual n^2 law.
- **Wavefront and inhomogeneity effects.** These are generally tied to beam geometry, cloud expansion, spatial inhomogeneity, and apparatus-specific configuration. They may grow with LMT, but they are not expected to produce the same joint dependence on n^2, T, and pre-split coherence preparation.
- **Readout and detection noise.** These affect the uncertainty floor more directly than they generate an intrinsic residual coherence burden.

The useful discriminator is therefore not total contrast loss by itself, but a residual component that survives ordinary technical mitigation and retains the joint scaling

$$D_{\mathrm{pred}} \propto T\, n^2\, k_{\mathrm{eff}}^2 \left(\frac{w_{\mathrm{coh}}}{\lambda_{\mathrm{coh},0}}\right)^2 ,$$

or, under the thermal identification, a quadratic dephasing coefficient that scales linearly with Θ.

6 Falsifiers

The note is only useful if it states what would count against the proposal. At least three clean falsifiers are available.

(i) **No quadratic residual.** If contrast loss is fully captured by linear or calibration-driven models across the accessible LMT range, the fixed-cell dephasing claim is disfavored.

(ii) **No preparation dependence.** If the residual loss does not track the pre-split coherence length $\lambda_{\mathrm{coh},0}$ or, under the thermal identification, the source temperature Θ in the predicted way, the fixed-cell closure is disfavored.

(iii) **Disappearance under pulse optimization.** If the entire apparent signature is removed by standard pulse-engineering improvements, then the effect is better interpreted as ordinary technical loss.

A positive result would therefore not merely be "contrast falls at high LMT," but rather:

> a residual dephasing component that survives ordinary technical mitigation and scales jointly with n^2, T, and smaller pre-split coherence length $\lambda_{\mathrm{coh},0}$, or thermally with higher source temperature Θ.

7 Experimental value of the test

The experimental value of this proposal lies in its asymmetry. A positive result would point to a nonstandard coherence burden tied to matter-wave transport itself. A null result would still be highly informative by constraining the fixed-cell closure and the specific route from the reduced transport theory to the laboratory forecast.

That is why MAGIS-100 is a particularly useful target. Even before ultimate design sensitivity is reached, the experiment can potentially probe the *shape* of the scaling law by controlled sweeps in LMT enhancement, preparation, and interrogation time.

8 Conclusion

The fixed-cell transport model predicts an additional dephasing exponent that can be stated in directly experimental variables:

$$D_{\mathrm{pred}} \propto T\, n^2\, k_{\mathrm{eff}}^2 \left(\frac{w_{\mathrm{coh}}}{\lambda_{\mathrm{coh},0}}\right)^2, \qquad \text{or thermally} \qquad D_{\mathrm{pred}} \propto T\, n^2\, k_{\mathrm{eff}}^2\, \Theta.$$

That prediction is sharper than the generic statement that large-momentum-transfer interferometers lose contrast at high pulse number. Its distinguishing content is the joint scaling with LMT enhancement, interrogation time, and smaller pre-split coherence length $\lambda_{\mathrm{coh},0}$, or thermally with higher source temperature Θ.

A recent primitive-cell closure analysis motivates the closure-informed choice $\eta = \pi$ used in the forecast band shown here, reducing the arbitrariness of the geometry side relative to a fully free shape sweep. It does not, however, by itself derive the laboratory mapping from source coherence length to operational cell area from first principles.

Accordingly, the most useful MAGIS-100 test is not a single headline measurement but a short discrimination program built from LMT sweeps, source-preparation sweeps, interrogation-time

sweeps, and ordinary pulse recalibration. The temperature sweep is especially valuable because it provides an additional derived control knob within the present bridge, rather than an auxiliary post hoc diagnostic. If a residual component with the proposed scaling survives those tests, it would provide direct evidence for the predicted matter-wave transport burden. If it does not, the fixed-cell bridge is correspondingly constrained. Either outcome is valuable.

References

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